Supervised Learning

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$$ } training set

Classification: task ⇒ to predict whether a car is a family car or not

$$x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \end{bmatrix} \begin{matrix} \rightarrow \text{price} \\ \rightarrow \text{engine power} \end{matrix}$$

price → [$10,000 – $1,000,000]

engine power → [1500 – 4000]

$$y_i = \begin{cases} +1 & \text{if } x_i \text{ is a family car} \\ -1 \text{ or } 0 & \text{otherwise} \end{cases}$$

data matrix

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \\ x_{N1} & x_{N2} \end{bmatrix} \quad x_1^T, x_2^T, \ldots, x_N^T$$

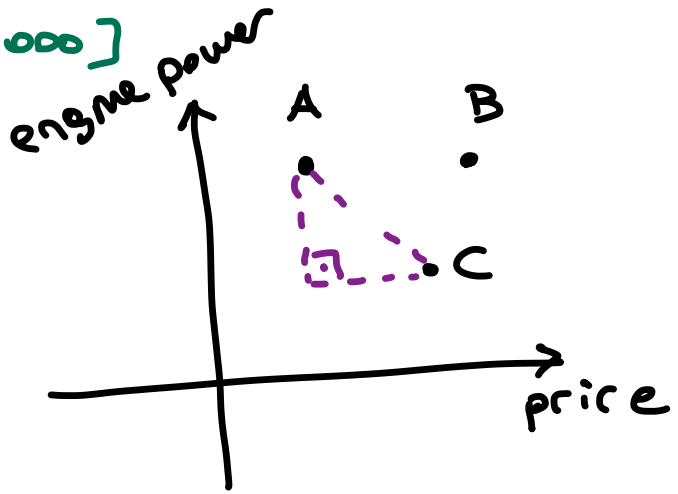
N × D

of data points

of features (D = 2)

label vector

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$ N × 1



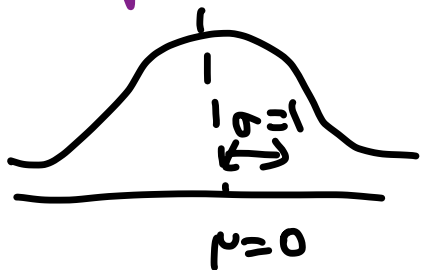
distance(A, C) $= \sqrt{\text{pricediff}^2 + \text{powerdiff}^2}$

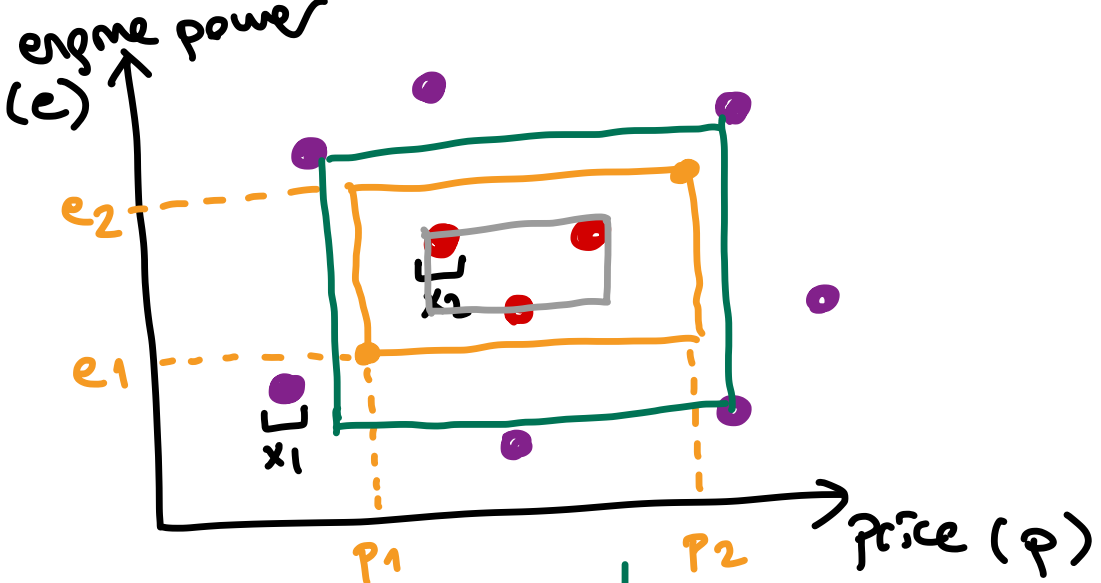
min-max scaling

$$\frac{x - \min}{\max - \min}$$

standard scaling

$$\frac{x - \mu}{\sigma}$$





Model Family

FAMILY OF RECTANGLES

● : family car (1)

● : other types of car (0)

$$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix} \quad x_2 = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix} \quad \dots$$

$$y_1 = [0] \quad y_2 = [1] \quad \dots$$

$$\theta = \{ p_1, p_2, e_1, e_2 \}$$

θ: model parameters

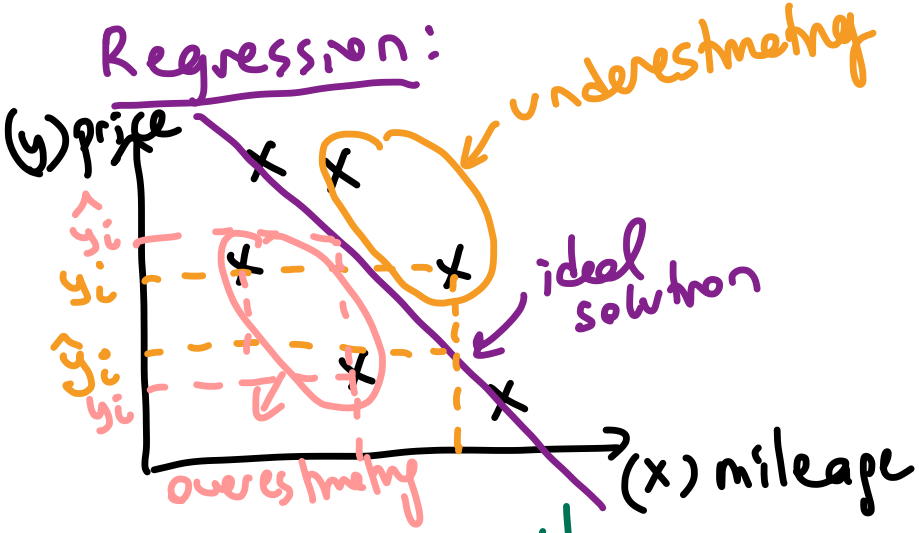
LEARNING $\Rightarrow$ Finding the best θ.

$$p_1^*, p_2^*, e_1^*, e_2^*$$

$$f(x_{N+1} \mid p_1^*, p_2^*, e_1^*, e_2^*) = \hat{y}_{N+1}$$

$$\begin{cases} \text{if } p_1^* \le x_{N+1,1} \le p_2^* \ \&\ e_1^* \le x_{N+1,2} \le e_2^* \\ \quad \hat{y}_{N+1} = 1 \quad \leftarrow \text{inside the rectangle} \\ \text{else} \\ \quad \hat{y}_{N+1} = 0 \quad \leftarrow \text{outside of the rectangle} \end{cases}$$

Regression:



Model Family

FAMILY OF LINES

w_0^*, w_1^*

data matrix X = (mileage)

$$X = \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{N1} \end{bmatrix}_{N \times D}$$

output/target vector y = (price)

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

$\Rightarrow \quad w_0 + w_1 . x \qquad \theta = \{w_0, w_1\}$

$$\hat{y}_1 = w_0^* + w_1^* x_{11}$$

$$\hat{y}_2 = w_0^* + w_1^* x_{21}$$

$$\hat{y}_N = w_0^* + w_1^* x_{N1}$$

$$e_1 = y_1 - \hat{y}_1$$

$$e_2 = y_2 - \hat{y}_2$$

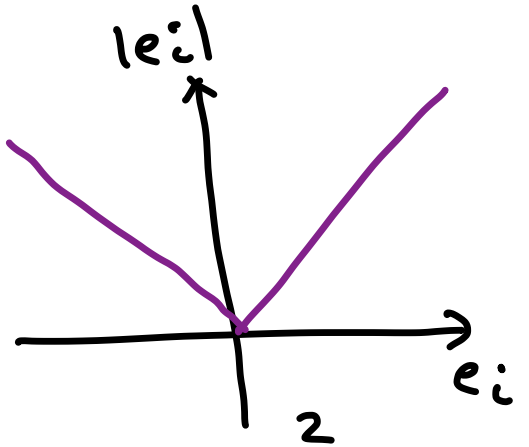
$$e_N = y_N - \hat{y}_N$$

x_{N+1}

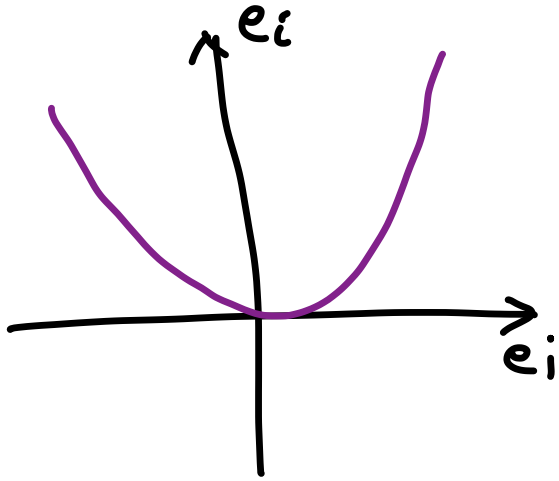
$$\hat{y}_{N+1} = w_0^* + w_1^* x_{N+1,1}$$

✗ minimize $\sum_{i=1}^{N} (y_i - \hat{y}_i) = \sum_{i=1}^{N} e_i$

✗ minimize $\sum_{i=1}^{N} |y_i - \hat{y}_i| = \sum_{i=1}^{N} |e_i|$



✓ minimize $\sum_{i=1}^{N} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{N} e_i^2$



minimize $\sum_{i=1}^{N} \left[y_i - \left[w_0 + w_1 \cdot x_{i1} \right] \right]^2$

(✓ y_i, ? w_0, ? w_1, ✓ x_{i1})

with respect to: w_0, w_1

$$\frac{\partial [f(a) + g(a)]}{\partial a} = \frac{\partial f(a)}{\partial a} + \frac{\partial g(a)}{\partial a}$$

$$Error(w_0, w_1 \mid x) = \sum_{i=1}^{N} (y_i - (w_0 + w_1 x_{i1}))^2$$

$$\frac{\partial Error}{\partial w_0} = \sum_{i=1}^{N} \frac{\partial [y_i - (w_0 + w_1 x_{i1})]^2}{\partial w_0}$$

$$= \sum_{i=1}^{N} 2 \cdot (y_i - (w_0 + w_1 x_{i1}))^1 \cdot (-1) = 0$$

$$\frac{\partial Error}{\partial w_1} = \sum_{i=1}^{N} \frac{\partial [y_i - (w_0 + w_1 x_{i1})]^2}{\partial w_1}$$

$$= \sum_{i=1}^{N} 2 \cdot (y_i - (w_0 + w_1 x_{i1}))^1 \cdot (-x_{i1}) = 0$$

Exercise#1: Solve for w_0^* and w_1^*.

$$w_1^* = \frac{\sum_{i=1}^{N}(x_{i1}y_i) - \left(\sum_{i=1}^{N} x_i/N\right)\left(\sum_{i=1}^{N} y_i/N\right).N}{\sum_{i=1}^{N} x_{i1}^2 - N\left[\sum_{i=1}^{N} x_{i1}/N\right]^2}$$

$$w_0^* = \left(\sum_{i=1}^{N} y_i/N\right) - w_1^*\left[\sum_{i=1}^{N} x_{i1}/N\right]$$